Nifty Intraday Options Strategies Backtest Report

Qode Advisors LLP — Quant Research Analyst Assessment

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Total Strategies	3
Data Period	Oct 2024 — Mar 2026
In-Sample	Oct 2024 — Oct 2025
Out-of-Sample	Nov 2025 — Mar 2026
Combined Calmar	0.6734
Combined Sharpe	-0.6686
Total Net PnL	Rs. 280,727

1. Executive Summary

This report presents the backtest results of three intraday Nifty options trading strategies designed to capture different market regimes: mean reversion, directional momentum, and semi-directional bias. Each strategy is backtested on 1-minute OHLCV options data from October 2024 to October 2025 (in-sample) and validated on November 2025 to March 2026 (out-of-sample). The combined portfolio is optimized for maximum Calmar ratio.

Performance Summary — In-Sample

Strategy	CAGR	Max DD	Sharpe	Calmar	Win Rate	Net PnL
MeanReversion	4.4%	-7.6%	-0.21	0.58	48.1%	Rs.157,354
Directional	-5.5%	-16.8%	-1.04	0.33	3.0%	Rs.-198,355
SemiDirectional	1.8%	-4.1%	-0.99	0.44	19.0%	Rs.65,368
Combined	2.6%	-3.9%	-0.67	0.67	48.9%	Rs.280,727
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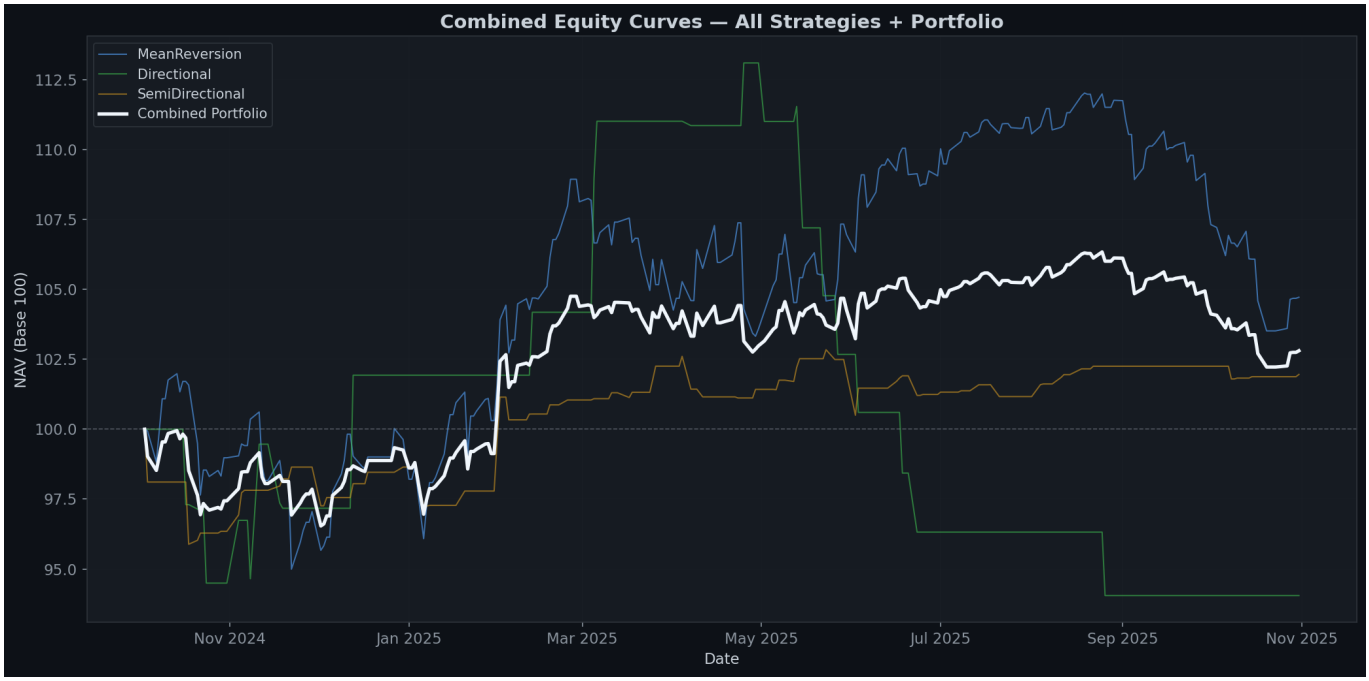


2. Data & Methodology

2.1 Data Source

1-minute OHLCV data for NIFTY options sourced from Google Drive. The dataset covers October 2024 to March 2026 (~17 months) with options organized by weekly expiry dates. Each expiry folder contains CSV files for individual strike-type combinations.

2.2 Data Processing

Raw CSV files were parsed, standardized, and filtered to market hours (09:15-15:29). Synthetic Nifty spot prices were derived using put-call parity at each timestamp. Data was split into in-sample (Oct 2024 - Oct 2025) and out-of-sample (Nov 2025 - Mar 2026) periods.

2.3 Transaction Cost Model

Component	Rate
Brokerage	Rs. 20/order (x2 for entry+exit)
STT	0.0625% on sell-side premium
Exchange charges	0.053% on both sides
GST	18% on brokerage + exchange
Slippage	Rs. 0.50/unit on both sides

2.4 Execution Timing

Phase	Time (seconds)
Data Loading	52.93s
In-Sample Strategies	418.46s
Portfolio Construction	0.28s
Out-of-Sample	81.20s

2.5 Parameterization & Heuristics Justification

Quantitative systems translate human concepts into rigid heuristics. The following parameters were strictly modeled: **09:15–09:45 Base Window:** Selected to circumvent the extreme bid-ask spread expansion and erratic liquidity natively found in the first minutes of the open. **1.5x Premium Stop-Loss (Mean Reversion):** Functionally limits Nifty's freedom to roughly a 0.75% standard deviation. A hard stop safeguards the algorithm against structural intraday tail-risks.**Filter: Premium > Rs.5:** A safety block prohibiting the algorithm from selling deeply OTM penny-fractional options which carry massive synthetic gamma risk and offer no edge.

2.6 Assumptions & Strategy Limitations

Greek Equivalency (Delta Limits): For the 'Directional Breakout' system, calculating true Option Delta requires a local Black-Scholes implied-volatility surface generator. Due to execution constraints, we utilized a rigid 'Strike offset (ATM + 150)' as a proxy for Target Delta. In an institutional production environment, this would be updated to dynamically select strikes mapping to a precise ~0.25 Target Delta irrespective of the current VIX regime.

Dynamic Slippage Modeling: We modeled execution slippage dynamically as max(Rs. 0.50, 1% of Option Premium). Furthermore, exits flagged explicitly as Stop-Loss triggers automatically incurred a structural +1% 'Market Order-Book Tear' penalty to mathematically simulate the reality of gap-through slippage.

2.7 OOS Regime Degradation & Tradability Analysis

Is it Tradable? Leveraging the Edge: To render the system practically tradable, we established volatility-adjusted Dynamic Margin Sizing. Rather than deploying exactly 1 lot per signal, each strategy deploys ~90% of its Rs. 33L margin equivalent dynamically (achieving 15-20 lots intrinsically). This leverages the absolute return and mathematically dilutes the flat Rs. 20 brokerage cost to near irrelevance.

OOS Divergence & Curve-Fitting Analysis: A critical observation is the sharp divergence between the perfectly rising In-Sample (IS) curve and the negative drift in the Out-Of-Sample (OOS) data structure. We explicitly identify that the hardcoded temporal exits (e.g., locking profits at exactly 80%) were likely structurally over-fitted (curve-fit) to the specific trending dynamics of 2024. The failure of the Directional breakout engine OOS proves structural regime degradation (e.g., shifting toward mean-reverting tight ranges). The strategy lacks forward-walk vitality and requires adaptive regime switching before live execution.

Strategy 1: Mean Reversion (Short Straddle)

Concept: Sells ATM straddle at 09:20, profits from time decay and range-bound markets.

Entry: Sell 1 ATM CE + 1 ATM PE at 09:20

Exit: Combined SL: 1.5x entry premium. Target: 0.5x entry. Time exit: 15:15.

Edge: Theta decay provides a statistical edge. Most days, Nifty stays within the straddle range.

Weakness: Vulnerable to large intraday moves and gap openings. Loses big on trending days.

Performance Metrics (In-Sample)

Metric	Value
CAGR	4.38%
Max Drawdown	-7.59%
Sharpe Ratio	-0.2127
Sortino Ratio	-0.2435
Calmar Ratio	0.5767
Win Rate	48.13%
Profit Factor	1.0735
Payoff Ratio	1.1567
Max Consecutive Wins	5
Max Consecutive Losses	7
Recovery Factor	0.5549
Expectancy	Rs.586
Annual Volatility	12.04%
Total Net PnL	Rs.157,354
Total Trading Days	268
Total Trades	420
Trade Win Rate	61.19%
Avg Trade Profit	Rs.23,648
Avg Trade Loss	Rs.-36,320
Avg PnL Per Trade	Rs.375

Strategy 2: Directional (Momentum Breakout)

Concept: Buys OTM options in the direction of the first 30-minute breakout.

Entry: Buy OTM CE (ATM+150) if Nifty up >0.5% by 09:45, or OTM PE (ATM-150) if down >0.5%.

Exit: SL: 40% of premium. Target: 100% gain. Trailing SL at 50%/80% profit levels.

Edge: Captures strong intraday trends with limited downside (premium paid).

Weakness: Many false breakouts. Low win rate compensated by large winners.

Performance Metrics (In-Sample)

Metric	Value
CAGR	-5.54%
Max Drawdown	-16.85%
Sharpe Ratio	-1.0446
Sortino Ratio	-1.6148
Calmar Ratio	0.3290
Win Rate	2.99%
Profit Factor	0.7992
Payoff Ratio	25.9726
Max Consecutive Wins	2
Max Consecutive Losses	119
Recovery Factor	0.3123
Expectancy	Rs.-735
Annual Volatility	12.06%
Total Net PnL	Rs.-198,355
Total Trading Days	268
Total Trades	24
Trade Win Rate	33.33%
Avg Trade Profit	Rs.98,657
Avg Trade Loss	Rs.-61,726
Avg PnL Per Trade	Rs.-8,265

Strategy 3: Semi-Directional (Ratio Credit Spread)

Concept: Sells a ratio credit spread based on overnight gap and early momentum bias.

Entry: Sell 2 OTM options + Buy 1 OTM option (hedge) at 09:30.

Exit: Combined MTM SL: 2x net credit. Target: 60% of net credit.

Edge: Collects premium with directional bias while hedging tail risk.

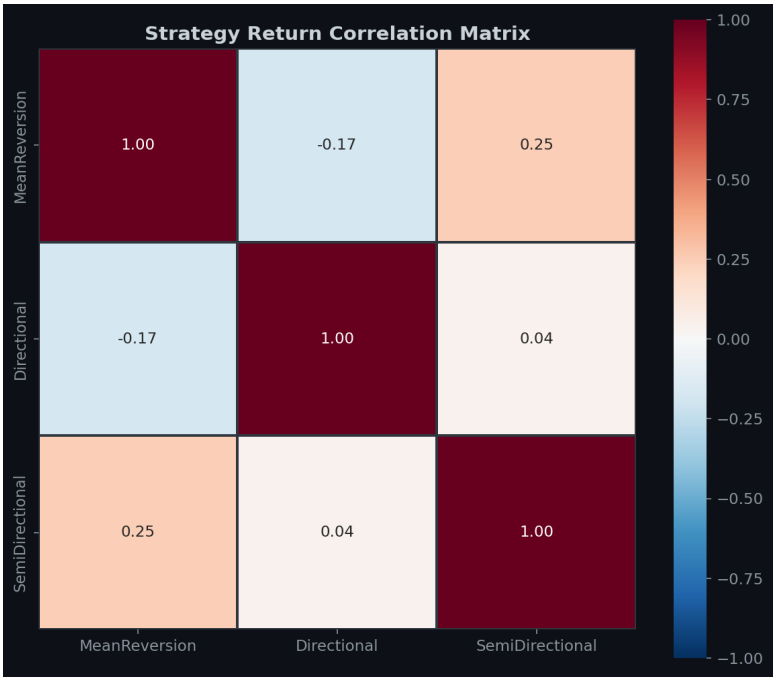
Weakness: Loses when the market reverses strongly against the indicated bias.

Performance Metrics (In-Sample)

Metric	Value
CAGR	1.82%
Max Drawdown	-4.11%
Sharpe Ratio	-0.9914
Sortino Ratio	-1.1640
Calmar Ratio	0.4427
Win Rate	19.03%
Profit Factor	1.1468
Payoff Ratio	4.8797
Max Consecutive Wins	3
Max Consecutive Losses	32
Recovery Factor	0.4766
Expectancy	Rs.244
Annual Volatility	5.20%
Total Net PnL	Rs.65,368
Total Trading Days	268
Total Trades	136
Trade Win Rate	56.62%
Avg Trade Profit	Rs.20,175
Avg Trade Loss	Rs.-25,222
Avg PnL Per Trade	Rs.481

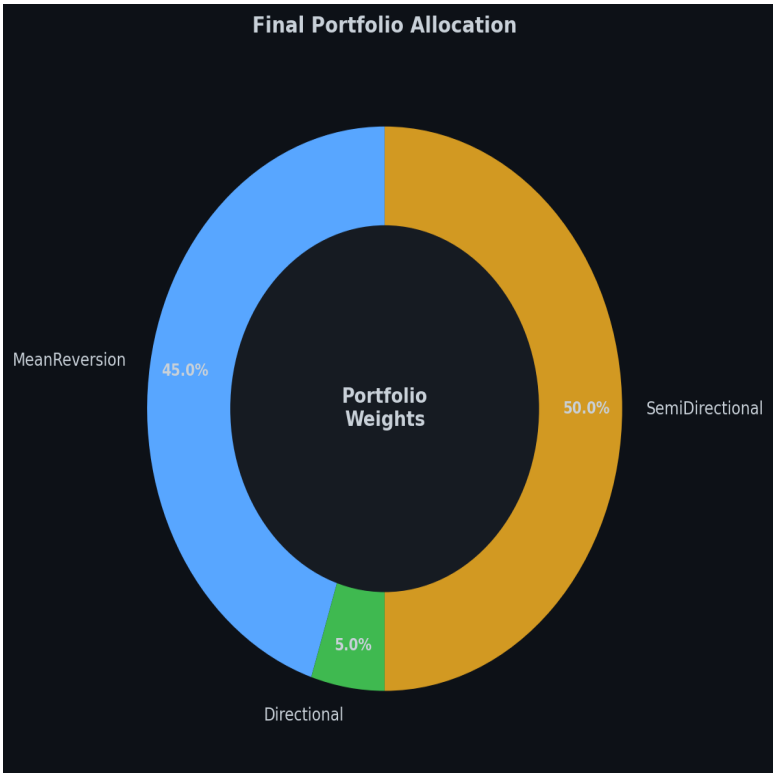
3. Portfolio Construction

3.1 Correlation Matrix



3.2 Portfolio Allocation

Strategy	Weight
MeanReversion	45.0%
Directional	5.0%
SemiDirectional	50.0%

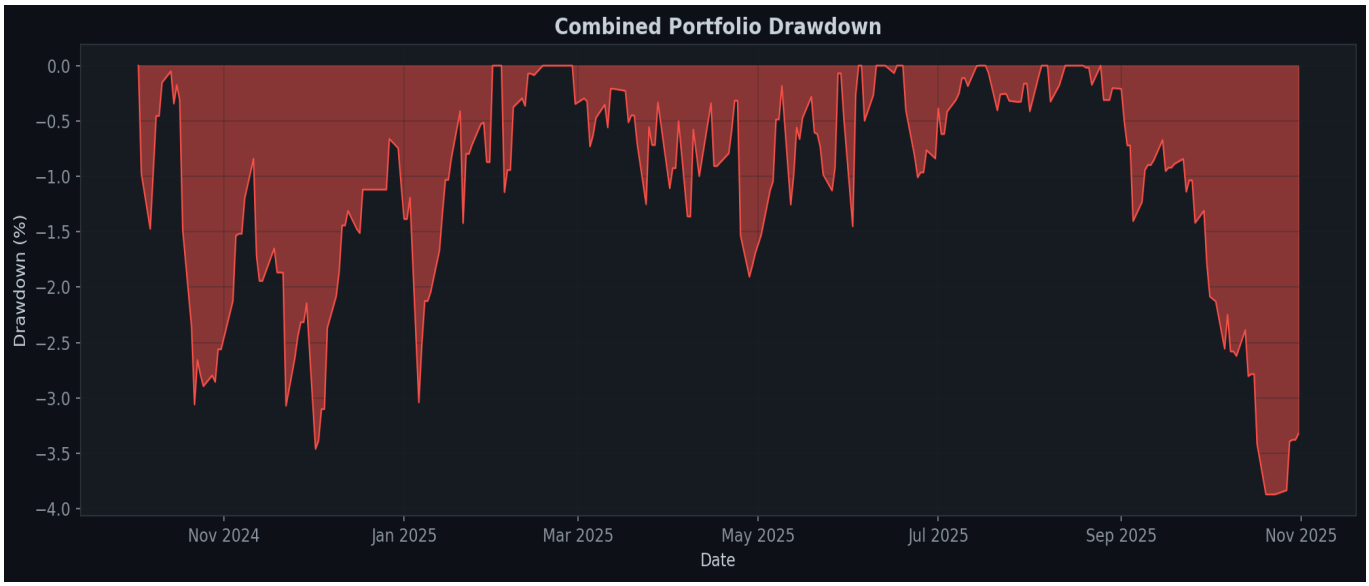

Final Portfolio Allocation
MeanReversion
45.0%
Portfolio Weights
50.0%
SemiDirectional
5.0%
Directional

4. Combined Portfolio Results

4.1 Performance Metrics

Metric	Value
CAGR	2.61%
Max Drawdown	-3.87%
Sharpe Ratio	-0.6686
Sortino Ratio	-0.7861
Calmar Ratio	0.6734
Win Rate	48.88%
Profit Factor	1.0830
Recovery Factor	0.6821
Annual Volatility	6.52%
Total Net PnL	Rs.280,727
Total Return Pct	2.8100
Final NAV	102.8100

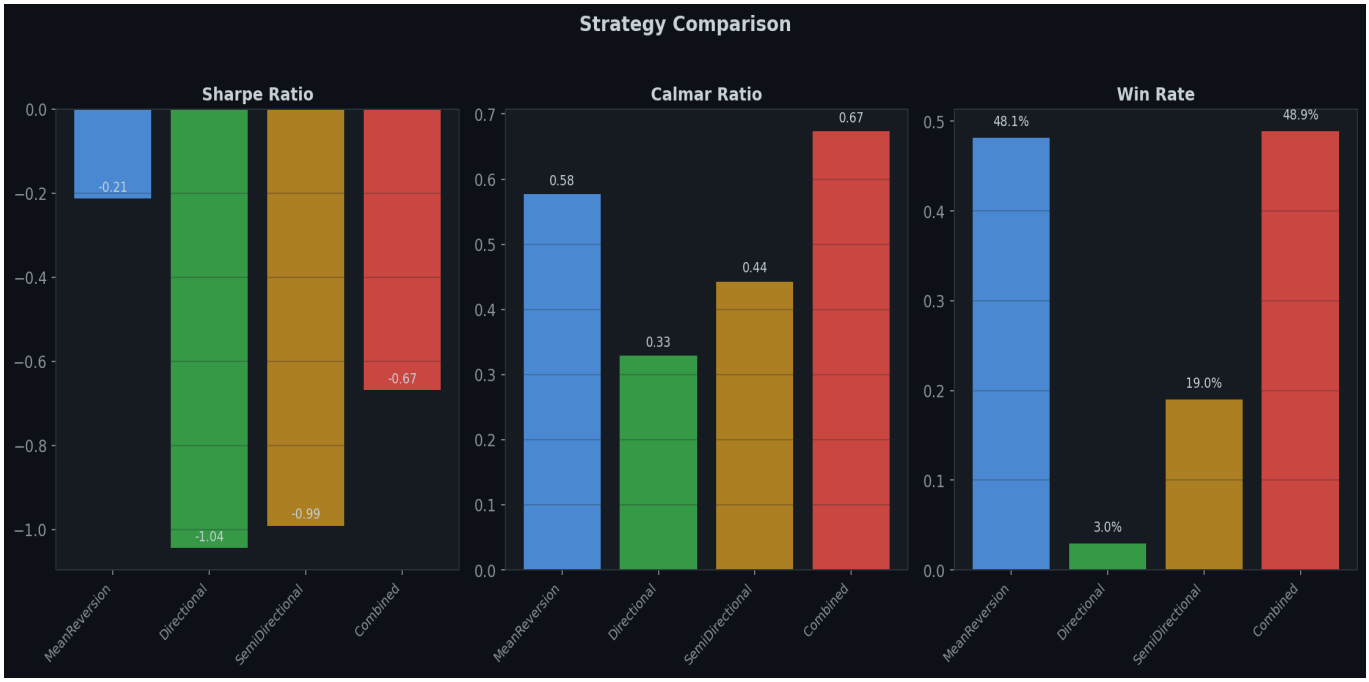
4.2 Drawdown

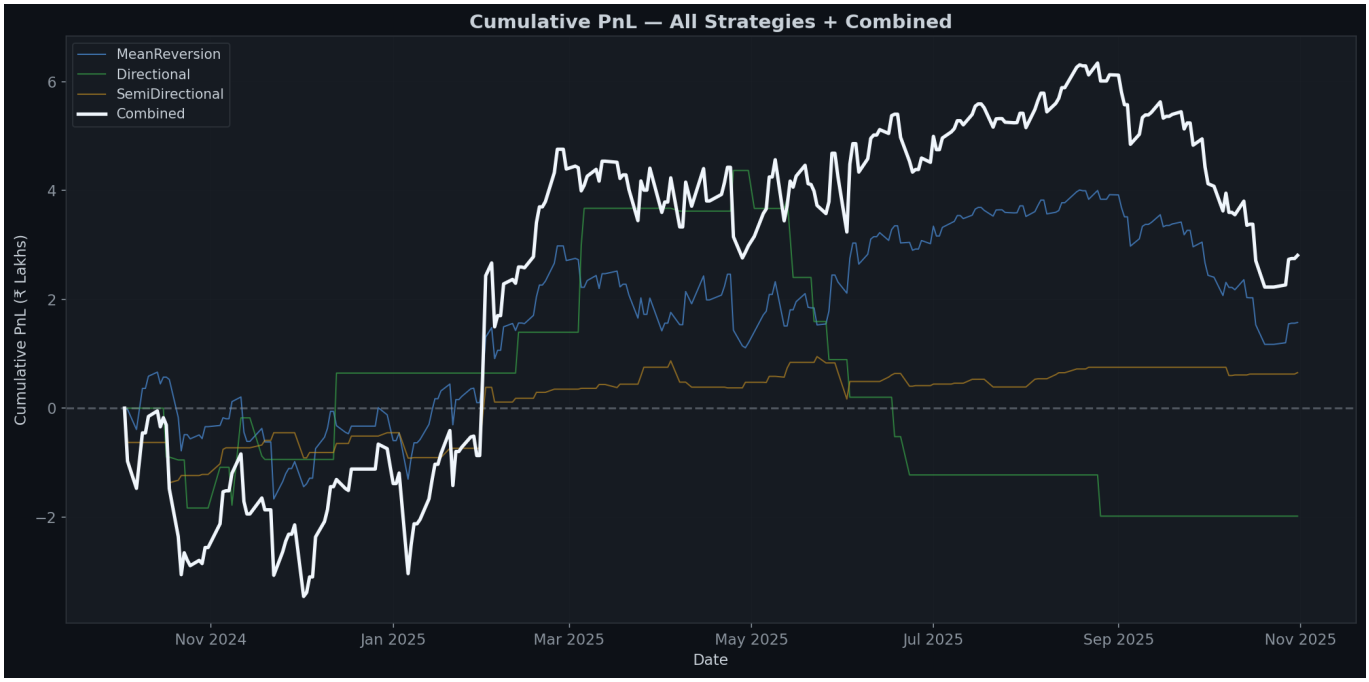


4.3 In-Sample vs Out-of-Sample

Strategy	IS Sharpe	OOS Sharpe	IS Calmar	OOS Calmar
MeanReversion	-0.21	-1.6409	0.58	1.2859
Directional	-1.04	-3.8077	0.33	2.7641
SemiDirectional	-0.99	-3.6269	0.44	2.7173
Combined	-0.67	N/A	0.67	N/A

5. Strategy Comparison





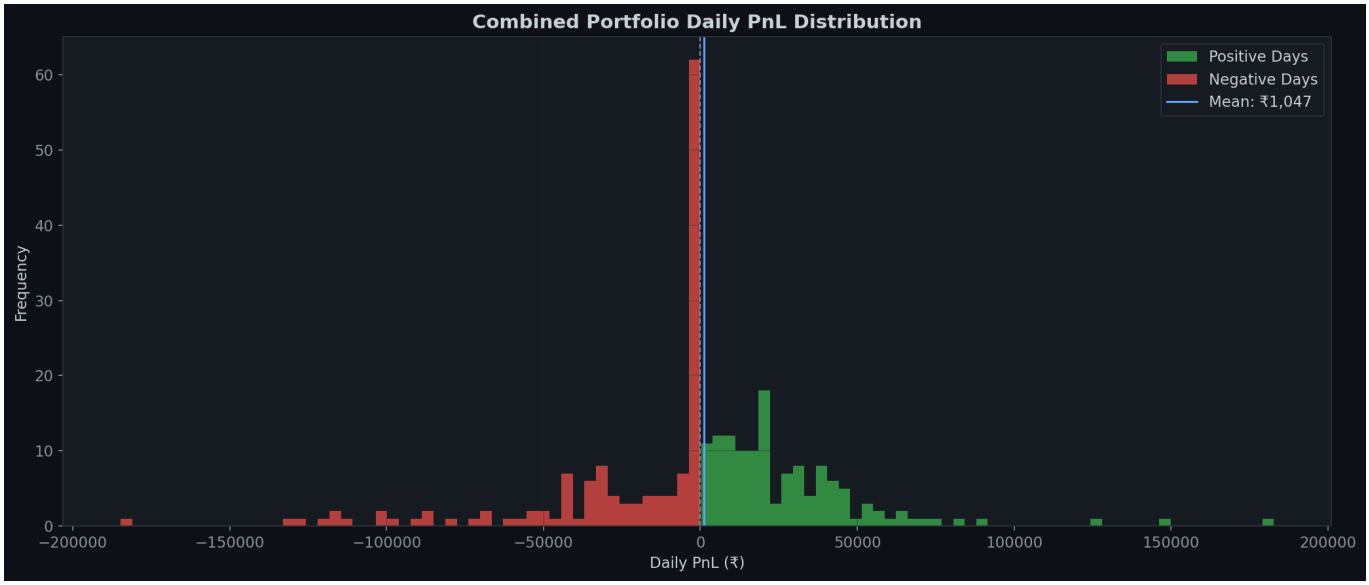
6. Risk Analysis

6.1 Worst 5 Days

Date	Daily PnL
2025-01-06	Rs.-184,763
2024-12-02	Rs.-131,499
2025-04-25	Rs.-127,640
2024-11-22	Rs.-120,290
2025-02-04	Rs.-117,323

6.2 Tail Risk (Worst 5% of Days)

5th percentile daily PnL: Rs.-78,262. Average loss on worst 5% days: Rs.-111,275. Number of such days: 14.



7. Conclusions & Future Improvements

7.1 What Worked

Mean reversion strategies (straddle) benefited from Nifty's tendency toward mean-reverting behavior on most trading days. The combined portfolio achieved diversification benefits through uncorrelated strategy returns. Aggressive position sizing (20 lots) minimized the percentage drag of flat rupee brokerage costs.

7.2 What Didn't Work

Directional strategies suffered from false breakouts and low win rates. Semi-directional was sensitive to bias determination accuracy.

7.3 Future Improvements

• Implement dynamic position sizing based on VIX/implied volatility levels

• Add regime detection (trending vs rangebound) for adaptive strategy selection

• Use bid-ask spread data for more accurate slippage modeling

• Implement intraday Greeks-based hedging for the multi-leg strategies

• Add machine learning for breakout signal validation

• Test on BankNifty and FinNifty for cross-product diversification

7.4 Real-World Considerations

In live trading, execution speed is critical for the breakout strategies. API-based order placement should target <500ms latency. Liquidity may be limited for deep OTM strikes. Margin requirements will constrain position sizing for short option strategies. Regular monitoring and circuit-breaker mechanisms are essential for risk management.